

EEE 554: Probability and Stochastic Processes

Practice Exam I - Detailed Solutions

Collaborative Solution Set

Spring 2026

Problem 1: Transformation of a Normal RV

If $x \sim \mathcal{N}[0, \sigma^2]$, find the PDF of $y = x^2$.

Solution. The PDF of x is $f_x(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/(2\sigma^2)}$. For $y > 0$, the CDF of y is:

$$F_y(y) = P(x^2 \leq y) = P(-\sqrt{y} \leq x \leq \sqrt{y}) = F_x(\sqrt{y}) - F_x(-\sqrt{y})$$

Differentiating with respect to y using the Chain Rule:

$$f_y(y) = \frac{d}{dy}[F_x(\sqrt{y}) - F_x(-\sqrt{y})] = f_x(\sqrt{y}) \frac{1}{2\sqrt{y}} + f_x(-\sqrt{y}) \frac{1}{2\sqrt{y}}$$

Since the normal PDF is symmetric about 0, $f_x(\sqrt{y}) = f_x(-\sqrt{y})$:

$$f_y(y) = \frac{1}{\sqrt{y}} f_x(\sqrt{y}) = \frac{1}{\sqrt{y}} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(\sqrt{y})^2/(2\sigma^2)} = \frac{1}{\sqrt{2\pi\sigma^2 y}} \exp\left\{-\frac{y}{2\sigma^2}\right\}, \quad y \geq 0$$

□

Problem 2: Probability of Independent Exponential RVs

RVs x and y are independent and exponentially distributed with parameters a and b , respectively. Calculate $P(\{x < y\})$.

Solution. The joint PDF is $f_{xy}(x, y) = f_x(x)f_y(y) = ae^{-ax} \cdot be^{-by}$ for $x, y > 0$.

$$\begin{aligned} P(x < y) &= \int_0^\infty \int_0^y ae^{-ax} be^{-by} dx dy \\ &= \int_0^\infty be^{-by} \left(\int_0^y ae^{-ax} dx \right) dy = \int_0^\infty be^{-by} (1 - e^{-ay}) dy \\ &= \int_0^\infty be^{-by} dy - \int_0^\infty be^{-(a+b)y} dy = 1 - \frac{b}{a+b} = \frac{a}{a+b} \end{aligned}$$

□

Problem 3: Order Statistics

RVs x_1, x_2, x_3 are independent and uniformly distributed on $[0, 1]$.

- (a) Define $y = \max(x_1, x_2, x_3)$. Find the PDF f_y and calculate $E[y]$.
- (b) Define $z = \min(x_1, x_2, x_3)$. Find the PDF f_z and $E[z]$.

Solution. For $x_i \sim U[0, 1]$, the CDF is $F_x(x) = x$ and the PDF is $f_x(x) = 1$ for $x \in [0, 1]$.

(a) $F_y(y) = P(x_1 \leq y, x_2 \leq y, x_3 \leq y) = [F_x(y)]^3 = y^3$.

$$f_y(y) = 3y^2, \quad 0 \leq y \leq 1; \quad E[y] = \int_0^1 y(3y^2) dy = \left[\frac{3y^4}{4} \right]_0^1 = \frac{3}{4}$$

(b) $P(z > z) = P(x_1 > z, x_2 > z, x_3 > z) = [1 - F_x(z)]^3 = (1 - z)^3$.

$$F_z(z) = 1 - (1 - z)^3 \implies f_z(z) = 3(1 - z)^2, \quad 0 \leq z \leq 1; \quad E[z] = \frac{1}{4}$$

□

Problem 4: Discrete RVs and Independence

- (a) $x \in \{-2, -1, 0, 1, 2\}$ with $P = 1/5$. $y = x^2 - 1$. Find statistics and independence.
- (b) $u \sim \mathcal{N}(\mu, \sigma^2)$, $v = u^2 - 1$. Find $E[v]$.
- (c) $z \in \{+1, -1\}$ with $P = 1/2$, $w = z^2 - 1$. Are z, w independent?

Solution. (a) $E[x] = 0$. $E[x^2] = \frac{1}{5}(4 + 1 + 0 + 1 + 4) = 2$, so $\text{var}(x) = 2$.

$y \in \{3, 0, -1, 0, 3\}$. $E[y] = 2 - 1 = 1$. $E[y^2] = \frac{1}{5}(3^2 \times 2 + 0^2 \times 2 + (-1)^2 \times 1) = 3.8$.
 $\text{var}(y) = 3.8 - 1^2 = 2.8$.

x, y are **not independent** because y is a function of x . For example, $P(x = 0, y = 3) = 0 \neq P(x = 0)P(y = 3)$.

(b) $E[v] = E[u^2] - 1 = (\text{var}(u) + E[u]^2) - 1 = \sigma^2 + \mu^2 - 1$.

(c) $w = z^2 - 1$. Since $z = \pm 1$, $z^2 = 1$ always, so $w = 0$ (constant). A constant is independent of any RV. □

Problem 5: Hypothesis Testing

Measurements x_k are i.i.d. $H_0 : x_k \sim U[0, 1]$. $H_1 : f_x(x) = 20x^{19}$ for $x \in [0, 1]$.

- (a) Binary quantization $b_k = 1$ if $x_k > T$. Find T for $P(b_k = 1|H_0) = 0.1$.
- (b) Global decision H_1 if $n = \sum b_k \geq 8$. Find P_d and P_f .

Solution. (a) $P(x_k > T|H_0) = 1 - T = 0.1 \implies T = 0.9$.

$P(b_k = 1|H_1) = \int_{0.9}^1 20x^{19} dx = [x^{20}]_{0.9}^1 = 1 - 0.9^{20} \approx 0.887$.

(b) $P_f = P(n \geq 8|H_0)$ where $n \sim \text{Binom}(10, 0.1)$.

$$P_f = \binom{10}{8}(0.1)^8(0.9)^2 + \binom{10}{9}(0.1)^9(0.9) + (0.1)^{10} \approx 3.736 \times 10^{-7}$$

$P_d = P(n \geq 8|H_1)$ where $n \sim \text{Binom}(10, 0.887) \approx 0.887$. □